SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Problem-Solution Fit

Date	1 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S)[CS] **• Small and medium-scale farmers.** **• Agricultural students.** **• Agricultural extension officers.** **• Individuals interested in smart farming.**	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** **• Limited technical knowledge .** **• Limited internet access in some areas .** **• Budget constraints.** **• Limited awareness of AI and machine learning tools.**	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Traditional farming advice: Easy to access but not always accurate. Agricultural experts: Reliable but not always available. Online resources: General information but not personalized.**
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** **• Difficulty selecting the right crop for soil conditions (Every cropping season).** **• Low crop yield due to incorrect crop selection (Frequent).** **• Lack of scientific decision support (Frequent) • Dependence on traditional farming practices (Very Frequent).**	**9. PROBLEM ROOT / CAUSE [RC]** **• Lack of data-driven crop selection methods • Limited understanding of soil nutrients • Changing climate and weather conditions • Limited access to agricultural technology**	**7. BEHAVIOR** + ITS INTENSITY **[BE]** **• Relies on previous farming experience.** **• Consults neighboring farmers .** **• Searches online occasionally.** **• Uses soil testing when available.**

2. TRIGGERS TO ACT [TR]

• Poor harvest in previous season.

• Soil test results available.

• Change in weather conditions.

• Need to increase agricultural productivity and income.

4. EMOTIONS

BEFORE / AFTER **[EM]**

Before: Confused, uncertain, worried about crop failure.
After: Confident, satisfied, and motivated to make data-driven farming decisions.

10. YOUR SOLUTION [SL]

OptiCrop is a machine learning-based web application that recommends the most suitable crop using soil nutrients (Nitrogen, Phosphorus, Potassium) and environmental parameters (temperature, humidity, pH, and rainfall). It provides instant crop recommendations through a simple Flask web interface, helping farmers improve productivity and make informed farming decisions.

8. CHANNELS OF BEHAVIOR [CH]

Online: Agricultural websites, YouTube, mobile applications, government agriculture portals.

Offline: Local agriculture officers, fellow farmers, soil testing centers, farming workshops.